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CROSS-DEVICE CORRELATION AND CONTEXTUAL INTERPRETATION METHODS

A. Overview

In various embodiments, the computing system correlates measurements from two or more nodes to improve interpretability, reduce ambiguity, and generate privacy-preserving outputs that reflect longitudinal patterns. Correlation may be performed between intimate-region nodes and context nodes, between external and intraluminal nodes, between geometry/baseline nodes and longitudinal nodes, and between any other combination of nodes.

B. Time Alignment and Segmentation

The computing system may align node data streams using timestamps, clock correction, synchronization beacons, session markers, and/or detected events. Measurements may be segmented into time windows comprising fixed windows, rolling windows, and event-triggered windows. In some embodiments, windows are labeled by context state, including sleep state, activity state, posture state, thermal state, and/or wear-integrity state.

C. Context Conditioning

In certain embodiments, the computing system conditions interpretation of intimate-region metrics based on physiologic context derived from an upper-body context node. Context conditioning may include adjusting thresholds, weighting confidence, or interpreting patterns differently depending on activity state, autonomic proxies, respiratory proxies, sleep proxies, thermal context, and/or contact-integrity state. For example, intimate-region pattern changes occurring during high-motion segments may be down-weighted, excluded, or flagged with reduced confidence.

D. Correlation Computation

Cross-device correlation may include one or more of: 1. Correlation scoring, including correlation coefficients, rank correlations, similarity scores, and/or learned similarity metrics. 2. Lag analysis, including identification of time offsets between changes in one node and changes in another node. 3. Sequence detection, including detection of ordered events, state transitions, or multi-signal motifs. 4. Co-variation analysis, including detection of joint changes across modalities. 5. Context-tagged correlation, including correlation computed within specific states such as sleep, rest, recovery, or activity. 6. Confidence scoring, including correlation confidence based on signal quality indices, wear validation, and artifact rejection. Correlation may be computed on raw signals, on derived features, on metrics, or combinations thereof. In some embodiments, correlation is computed using a rules-based approach; in other embodiments correlation is computed using statistical models, machine learning models, or personalized models.

E. Baseline and Geometry Normalization

In certain embodiments, baseline acquisition results and/or geometry-derived metrics are used to normalize and interpret longitudinal patterns. For example, outputs from a geometry measurement node may provide reference parameters that improve comparability across sessions. Outputs from an intraluminal compliance node may provide individualized response parameters used to scale, normalize, or interpret other node outputs.

F. Output Generation

The computing system may output correlation results using privacy-preserving representations comprising indices, zones, ranges, curves, and confidence indicators. Correlation outputs may include an alignment score, synchrony score, stability score, lag signature, and/or context-tagged correlation view. In certain embodiments, the system presents correlation outputs as within-user views across time windows and as between-node views across device combinations.

G. Variations

Correlation may be performed locally on a mobile device, on-node, in a cloud environment, or distributed across components. Correlation may be computed in real time, near real time, or offline. In certain embodiments, correlation methods are updated over time as additional user data becomes available, while maintaining privacy-preserving output formats.